

## 1. What is Type Casting?

**Type casting** is the process of converting a value from one data type to another. It helps make data compatible for different operations. Python supports both automatic and manual type conversion. Type casting improves flexibility and helps prevent type-related errors.

## 2. What is the Difference Between Implicit and Explicit Type Casting?

**Implicit type casting** is performed automatically by Python when there is no risk of data loss. **Explicit type casting** is performed manually by the programmer using functions like `int()`, `float()`, and `str()`. Implicit conversion requires no user intervention, whereas explicit conversion is done when needed. Both are used to convert data from one type to another.

## 3. What is the `type()` Function?

The **`type()`** function is used to determine the data type of a variable or object. It returns the class or type of the given value. It is commonly used for debugging and verifying data types during program execution. This function helps programmers understand what type of data they are working with.

## 4. Why is Type Casting Needed?

**Type casting** is needed to convert data into a compatible type before performing operations. It helps avoid type mismatch errors and ensures calculations or other operations work correctly. It also allows different data types to interact when required. This makes programs more flexible and reliable.

## 5. What are the Built-in Functions Used for Type Casting?

Python provides several built-in functions for type casting, such as **`int()`**, **`float()`**, **`str()`**, **`bool()`**, **`list()`**, **`tuple()`**, **`set()`**, and **`dict()`**. These functions convert values from one data type to another. They are widely used in Python programs to process different types of data. They make data conversion simple and efficient.